EXPERIMENTAL RESULTS AND COMPARISON

FreeMCG counterfactual generation: reverse-diffusion step-count ablation

Study: Compare the original 100-step reverse-diffusion setting with reduced settings of 50 and 25 steps.

Source materials

1. Original research paper: *Derivative-Free Diffusion Manifold-Constrained Gradient for Unified XAI* (CVPR 2025).
2. Executed Phase 2 Jupyter notebook supplied with this request.

Important distinction

Published paper results and this project's own notebook results are reported in separate sections. They are produced using different experimental protocols and must not be treated as directly comparable measurements.

Prepared from the uploaded research paper and executed notebook. Metrics and plots below are transcribed/extracted from the notebook outputs; no additional model runs were performed while preparing this PDF.

1. Executive summary

This project studies whether reducing the number of reverse-diffusion steps in FreeMCG can reduce counterfactual-generation runtime, and what happens to target-class success and similarity-related metrics when the step count is reduced.

Experiment: three correctly classified ImageNet validation images (cheetah, rhinoceros beetle and tabby cat), each evaluated with a shared target class of ladybug and three step settings (100, 50 and 25). This produces nine image/step conditions. Each condition uses 100 particles.

Main result: mean runtime fell from 35.45 minutes at 100 steps to 17.72 minutes at 50 steps and 8.86 minutes at 25 steps (approximately 50% and 75% faster). However, mean target-class probability fell substantially, and the mean flip rate remained very low. Therefore, the experiment demonstrates a runtime reduction, but does **not** demonstrate that the reduced-step settings preserve counterfactual target success.

The notebook includes one image per source class only. Its aggregate values are descriptive summaries across these three selected images, not statistically representative estimates for the full classes.

2. Research question and proposed improvement

Research question: How much can reverse-diffusion steps be reduced in FreeMCG before target-class success and source-image similarity deteriorate substantially?

Baseline: 100 reverse-diffusion steps, following the original FreeMCG reverse-diffusion counterfactual approach used in the notebook.
Proposed ablation: reduce the reverse-diffusion step count to 50 and 25 while holding the other listed experiment settings fixed.

This is a step-count ablation of the existing method, not a new counterfactual-generation algorithm.

3. Evidence from the original research paper

The following are extracts of the uploaded original paper. The page numbers below refer to the printed page numbers visible in the paper.

Original paper, printed page 6: Figure 4 compares input images and counterfactual outputs from DVCE, LDCE and FreeMCG. The figure includes examples such as Cheetah → Tiger, Mashed potato → Carbonara, and Lotion → Face powder. The accompanying text explains that reverse diffusion can be slower but can produce higher-fidelity results.

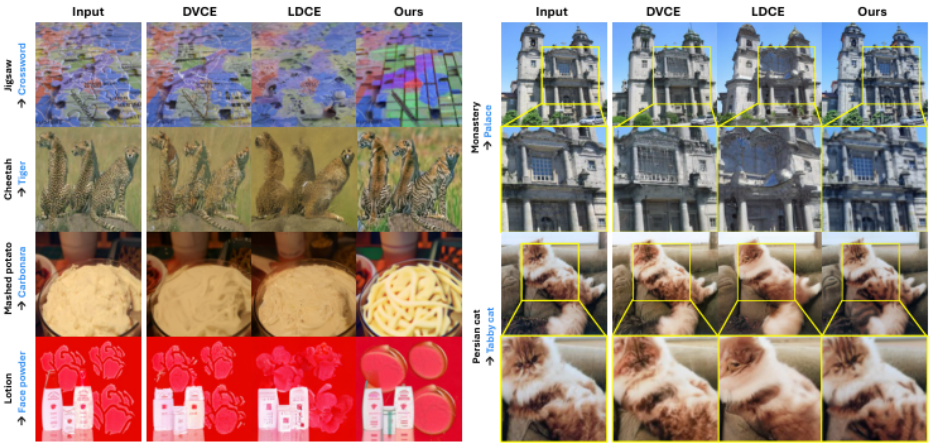


Figure 4. **Counterfactual Generation.** Images on the leftmost column are given as input, and the counterfactual for the target class (shown in blue) are generated using each method. Baseline methods generate counterfactuals by perturbing the image towards the direction of a given classification neural network's gradients. Not only are these methods not applicable to situations without access to model weights, gradients almost always point away from the image manifold and into the ambient space, leading to imperceptible changes (i.e., adversarial attack) or changes that produce unnatural, off-manifold, images. FreeMCG on the other hand produces semantically meaningful changes while keeping the image on-manifold.

tively apply gradient ascent with FreeMCG, i.e.

$$\boldsymbol{x}_{i+1} = \boldsymbol{x}_i + \alpha \boldsymbol{g}_{\text{Free}}(\boldsymbol{x}_i) + \beta(\boldsymbol{x} - \boldsymbol{x}_i), \tag{13}$$

where α is the step size weighting the FreeMCG, and β weights the proximal regularization to the initial point $\boldsymbol{x}$. Notice that the implementation of FreeMCG for counterfactual explanation differs from that of feature attribution, as we have a different target class c' from c. For instance, consider the example in (12), but the target class is now 2. Then, in the earlier stage of gradient ascent, we have

$$\begin{aligned} \boldsymbol{e}_c &= [0,1,0,0]^\top, \boldsymbol{p} = [0.7, 0.1, 0.1, 0.1]^\top \\ \boldsymbol{e}_c - \boldsymbol{p} &= [-0.7, 0.9, -0.1, -0.1]^\top. \end{aligned} \tag{14}$$

In other words, the components correlated with the original class c is strongly negatively pulled down, whereas the components correlated with the target class c' are strongly positively pushed up.

FreeMCG meets reverse diffusion. While iterative gradient ascents using FreeMCG in (13) gives us a fast way to generate counterfactual explanations, we find that this process is sensitive to hyperparameters and produces suboptimal results. To fully leverage the generative nature of diffusion models, we instead propose a variant that utilizes reverse diffusion, as is done in other methods that use diffusion models for counterfactual explanation [4, 15, 21]. Specifically, we use an SDEdit [28]-like approach, where we forward diffuse our initial sample up to time t' and run reverse guided sampling in (13) to the Tweedie denoised estimates. The algorithm is presented in Alg. 3. Notice that we do not incorporate the Jacobian of $\boldsymbol{\epsilon}_\theta$, which is typically done in methods like [8, 45], and simply resort to the gradients with respect to the Tweedie estimates [9] to save computation. For further details, see Appendix E. This approach may be slower as it requires reverse diffusion, but produces higher-fidelity results (Fig. 4). Therefore, all experiments on counterfactual explanation with FreeMCG use this reverse diffusion approach unless specified otherwise.

4. Experiments

We demonstrate our method from both feature attribution and counterfactual perspectives. To evaluate generalization and domain-specific performance, we compare our method on ImageNet [11] and MIMIC CXR [23] dataset, using images with a resolution of 256×256. We use regular pretrained classifier models with the ResNet-50 [17] and unconditional diffusion models [12] trained on each dataset. Experimental details are provided in Appendix F.

4.1. Evaluation protocol

Counterfactuals. An ideal counterfactual makes semantically meaningful, human-interpretable yet minimal changes to the input image so that it is classified as the target class by the model of interest. This allows the user to gain visual

Extract from the original research paper, printed page 6. Figure 4 and Section 3.2 are relevant to the counterfactual-generation method and its reverse-diffusion variant.

3.1 Original paper: evaluation protocol (printed page 7)

The paper evaluates counterfactuals using multiple criteria: target-class flip rate, human-perceived change, L2 distance, perceived similarity, FID and perceived realism. It notes that these metrics should be interpreted together rather than separately. The paper's ImageNet experiment uses 1,000 generated counterfactuals (one from each validation class) and selects a target category based on WordNet proximity to the input category.

insight into the decision boundary of the model. As counterfactual generation aims to create an example at a point on or near the decision boundary closest to the input image (Fig 2), the final generated counterfactual must be anchored on the original input image and should not be an unconditionally generated image of the target class. Therefore, evaluation of counterfactuals involves the following interrelated criteria:

1. **Meaningful change towards target class**
   (a) **Flip rate:** Is the generated counterfactual classified as the intended target class by the model of interest?
   (b) **Perceived change:** Does the generated counterfactual contain semantically meaningful changes to a human perceiver? (i.e. can it be distinguished from an *adversarial attack*?)
2. **Similarity to the input**
   (a) **L2 distance:** Are generated counterfactuals close to the input images in terms of distance?
   (b) **Perceived similarity:** Do generated counterfactuals and input appear similar to a human perceiver?
3. **Realism of generated image**
   (a) **Frechet Inception Distance (FID) [18]:** Is the distribution of the generated counterfactual images similar to distribution of original input images?
   (b) **Perceived realism:** Do the images appear realistic to a human perceiver?

A crucial aspect about these metrics is that they must be read and understood in concert, rather than disparately as separate measures. In particular, note that an *adversarial attack* [10, 16], which flips the decision of the classification neural network but does not introduce changes to the input image that are meaningful to a human perceiver (which are meaningless as explainability tools), can speciously score well on aspects such as similarity or realism. Assessment of similarity and realism scores must thus be predicated on the fact that there are meaningful changes that have been made towards the target class and viewed in proportion to that change: $\texttt{Change} \times (\texttt{Similarity} + \texttt{Realism})$.

Flip rate, L2 distance, and FID are computed using 1K generated counterfactuals (one from each class of the ImageNet validation set). Following the experimentation scheme in LDCE [15], we select the category closest to the input image based on the WordNet [30] hierarchy as the target class for counterfactual generation. Perceived change, perceived similarity, and perceived realism were measured through a human user study (Appendix G). Results are provided in Table 1.

We select DVCE [4] and LDCE [15] as baselines, leveraging diffusion models to generate counterfactuals. Note that these baseline methods require access to the model weights while FreeMCG does not. In addition, while the proposed method in DVCE includes a version with cone projection of model weights to the weights of an adversarially robust classifier, we use only a regular non-robust classifier for comparison as we are assuming lack of access to the training data or adversarially robust weights.

Feature attribution. Feature attribution maps can be evaluated by observing the change in model predictions whilst removing the pixels in order of purported importance from the attribution. We compare our approach with existing gradient-based attribution methods, including the vanilla gradient [43], Integrated Gradients [42] and InputXGradient [39]. Shapley additive explanation (SHAP) [27] is an effective model-agnostic feature attribution method that determines the importance of each feature by assessing the change in classifier predictions after removing a certain feature from possible combinations of features that make up a data point. For images, this amounts to dividing an image into clusters of pixels ('superpixels') and determining the importance of each superpixel by observing the effect of their removal. Since the size of the superpixels in the SHAP is usually very large, the SHAP maps are not usually as perceptually informative as gradient-based attribution methods.

As for the metric for the feature attribution, Remove and Debias (ROAD) [34] is a removal-based evaluation metric that minimizes information leakage from the shape of the pixel removal mask using noisy linear imputation. As such, ROAD is ideally suited with SHAP method, so under the assessment by removal-based metrics such as ROAD, SHAP feature attributions (Fig. 5) can be used as a sort of 'gold standard' to determine which attributions are most faithful to the model being explained.

4.2. Results

Counterfactual generation. Counterfactual generation results are shown in Fig 4. The generation results show that FreeMCG is able to robustly approximate the direction of the gradient and push the image towards the decision boundary of the classifier. Quantitative results and human user study results for the criteria outlined in Section 4.1 are shown in Table 1. Because baseline methods make use of model gradients while FreeMCG does not, the flip rate of the model's decision is lower for FreeMCG than gradient-based baselines. However, FreeMCG yields by far the most human-perceived changes (Tab. 1 and Fig. 4). In fact, FreeMCG-based counterfactuals often contain informative changes without changing the decision of the classifier while gradient-based baselines commonly produce counterfactuals that *fool* the classifier to predict a different class but do not introduce semantically meaningful changes. Gradient-based methods are thus closer to *adversarial attacks*, while FreeMCG better serves the purpose of counterfactual generation and produces interpretable, on-manifold changes towards the target class.

Feature attribution. Feature attribution maps from rep-

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Extract from the original research paper, printed page 7. See Section 4.1 for evaluation criteria and the paragraph describing the 1,000-image protocol and WordNet-based target selection.

3.2 Original paper: published quantitative results (printed page 8)

Table 1 in the paper reports the following published results. These values belong to the original authors and are **not** results generated by this project.

Method	Flip rate ↑	Perceived change ↑	L2 ↓	Perceived similarity ↑	FID ↓	Perceived realism ↑
DVCE	85.8	2.38	995.92	3.26	71.38	2.77
LDCE	61.7	2.59	1351.39	2.87	63.04	2.59
FreeMCG (paper)	51.5	3.74	1001.02	3.52	66.89	3.13

Published results transcribed from Table 1, original paper printed page 8. The arrows indicate the preferred direction of each metric.

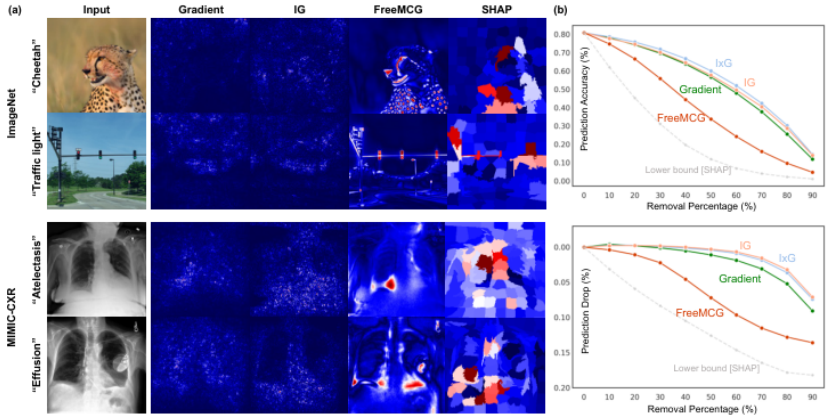


Figure 5. (a) Feature attribution maps obtained using each method, (b) Corresponding ROAD [34] evaluation results. Steeper decline is better, and SHAP is given as reference for the lower bound of this decline. For ImageNet, ROAD is calculated as described in [34]. For MIMIC chest X-ray data, results for decline in prediction accuracy was noisy due to the lower starting accuracy of the classifier, low number of classes, and severe data imbalance between classes; therefore, we instead measure the drop in prediction for the originally predicted class. The principle remains the same in that features (pixels) are removed in order of purposed importance from each feature attribution map to observe the change in classifier predictions.

	Change		Similarity		Realism	
	Flip Rate ↑	Perceived change ↑	L2 ↓	Perceived similarity ↑	FID ↓	Perceived realism ↑
DVCE [4]	**85.8**	2.38	**995.92**	3.26	71.38	2.77
LDCE [15]	61.7	2.59	1351.39	2.87	**63.04**	2.59
Ours	51.5	**3.74**	1001.02	**3.52**	66.89	**3.13**

Table 1. Quantitative (Flip Rate, L2, FID) and human user study (perceived change, perceived similarity, perceived realism) results.

resentative gradient-based methods for both ImageNet and MIMIC-CXR are shown in Fig. 5(a). ROAD evaluation results are also shown in Fig. 5(b). We see that baseline gradient methods provide feature attributions that are similar to adversarial attacks, hence hardly interpretable. SHAP maps provide less informative patch-like artifact. FreeMCG, on the other hand, produces high quality and interpretable attributions that are similar to SHAP, but retaining the high-frequency details as it does not rely on arbitrary pixel clustering. In ROAD evaluation, we see that FreeMCG achieves the steepest decline in model prediction, corresponding to better feature attributions, closer to the lower-bound computed by SHAP.

5. Conclusion

We developed Derivative-Free Diffusion Mainfold Constrained Gradients (FreeMCG), an approximation of a neural network's gradient tangent to the data manifold, obtained *without* access to the model weights or its real gradients. Previous explainability literature on gradient-based feature attribution and counterfactual generation have remained largely separate because the gradient could not be readily employed for counterfactual generation as it lies off the manifold and produces adversarial attacks rather than counterfactuals. With FreeMCG, one can obtain gradient directions on the image manifold that produce semantically meaningful changes and thus counterfactual explanations. We demonstrate through experiments that FreeMCG not only enables state-of-the-art counterfactual generation but also serves as highly informative feature attributions, serving as a unified framework for both explainability methods, all without access to model weights.

Original paper extract, printed page 8, containing Figure 5 and Table 1.

The paper's FreeMCG flip rate is 51.5% on its 1,000-counterfactual evaluation protocol. The paper also reports a higher human-perceived change score than the two baselines. These results are context for the method, not a direct benchmark against the small three-image experiment below.

4. Results obtained in our implementation

4.1 Dataset, model and selected images

The notebook selects one ImageNet validation image per source category and verifies that ResNet-50 predicts the requested source class for each selected image. The selected filenames and model predictions are:

Source category	ImageNet validation filename	Predicted source label	Target
Cheetah	ILSVRC2012_val_00005006.JPEG	cheetah	ladybug
Rhinoceros beetle	ILSVRC2012_val_00003191.JPEG	rhinoceros beetle	ladybug
Tabby cat	ILSVRC2012_val_00016389.JPEG	tabby	ladybug

4.2 Why is ladybug the target for all three images?

The notebook explicitly defines one shared target class: **ladybug**. The original predictions remain cheetah, rhinoceros beetle and tabby cat; ladybug is the desired class for the generated counterfactuals, not the initial classification of the three input images.

This is a deliberate controlled-design choice: the target class is held constant while the source image changes, and the reverse-diffusion step count is varied. Keeping the target fixed avoids adding target-category variation as another experimental factor. It allows this experiment to ask how 100, 50 and 25 steps behave for these three selected source-to-ladybug transformations.

Important qualification: the original paper's main ImageNet evaluation selected a target category based on WordNet proximity to each input class (printed page 7). The notebook's shared-ladybug design is therefore a different, narrower protocol. It is not inherently incorrect, but its conclusions apply to these three source-to-ladybug cases and should not be generalised to all possible target classes. In the original paper's qualitative examples, rhinoceros beetle → ladybug is shown, whereas cheetah → tiger and tabby cat → Persian cat are examples of different target choices.

4.3 Fixed settings and varied parameter

Varied: reverse-diffusion steps = 100, 50, 25.
Held fixed: K = 100 particles; strength = 0.4; start timestep = 400; FreeMCG step size $\alpha = 0.2$; proximal weight $\beta = 0.02$; $\eta = 0.0$; image, target class and seed remain fixed within each image's three step-count conditions. The notebook also caps execution micro-batches at 5 for memory safety and batches classifier calls; its equivalence check reports maximum absolute logit difference 0.0 and PASSED.

5. Per-image experimental results

All nine conditions from the executed notebook are listed below. Runtime is in minutes; flip rate is the percentage of the 100 generated particles classified as the target class; mean target probability is averaged over the generated particles; mean L2 is the reported average distance from the source.

Source image	Steps	Runtime (min)	Flip rate (%)	Mean target prob.	Mean L2
Cheetah	100	35.49	0.0	0.008333	115.41
Cheetah	50	17.74	0.0	0.000922	74.96
Cheetah	25	8.87	0.0	0.000642	58.48
Rhinoceros beetle	100	35.41	4.0	0.039847	80.89
Rhinoceros beetle	50	17.69	0.0	0.006037	84.74
Rhinoceros beetle	25	8.84	1.0	0.005596	71.55
Tabby cat	100	35.44	0.0	0.002560	94.68
Tabby cat	50	17.72	0.0	0.000711	81.40
Tabby cat	25	8.87	0.0	0.000572	65.76

Observation: the rhinoceros-beetle source image is the only one with non-zero flip rate in the tested conditions: 4% at 100 steps and 1% at 25 steps. The cheetah and tabby-cat cases have 0% flips at all three settings. Thus, target-class success is weak overall in this particular experiment.

6. Aggregate comparison: 100-step baseline vs reduced-step variants

Values below are mean ± standard deviation across the three selected image-level results. Because there is one image per source class, the standard deviation describes variation among these three selected images only; it is not a within-class estimate.

Steps	Runtime mean ± SD (min)	Flip rate mean ± SD (%)	Target probability mean ± SD	Mean L2 mean ± SD
100 (baseline)	35.445 ± 0.039	1.333 ± 2.309	0.016914 ± 0.020070	96.994 ± 17.379
50	17.718 ± 0.022	0.000 ± 0.000	0.002557 ± 0.003016	80.367 ± 4.968
25	8.860 ± 0.016	0.333 ± 0.577	0.002270 ± 0.002881	65.260 ± 6.547

Runtime reduction: 50 steps reduced mean runtime by approximately 50.0% relative to 100 steps; 25 steps reduced it by approximately 75.0%.
Target probability: mean target probability decreased from 0.016914 at 100 steps to 0.002557 at 50 steps and 0.002270 at 25 steps.
Flip rate: mean flip rate was 1.333% at 100 steps, 0% at 50 steps and 0.333% at 25 steps.
L2: mean L2 decreased as steps were reduced, but a lower L2 alone does not prove a better counterfactual; it may simply indicate less movement away from the source.

7. Notebook-generated aggregate graphs

The following figures are the plots displayed by the final graphing cell of the executed notebook. They are reproduced from the notebook outputs rather than redrawn from a separate run.

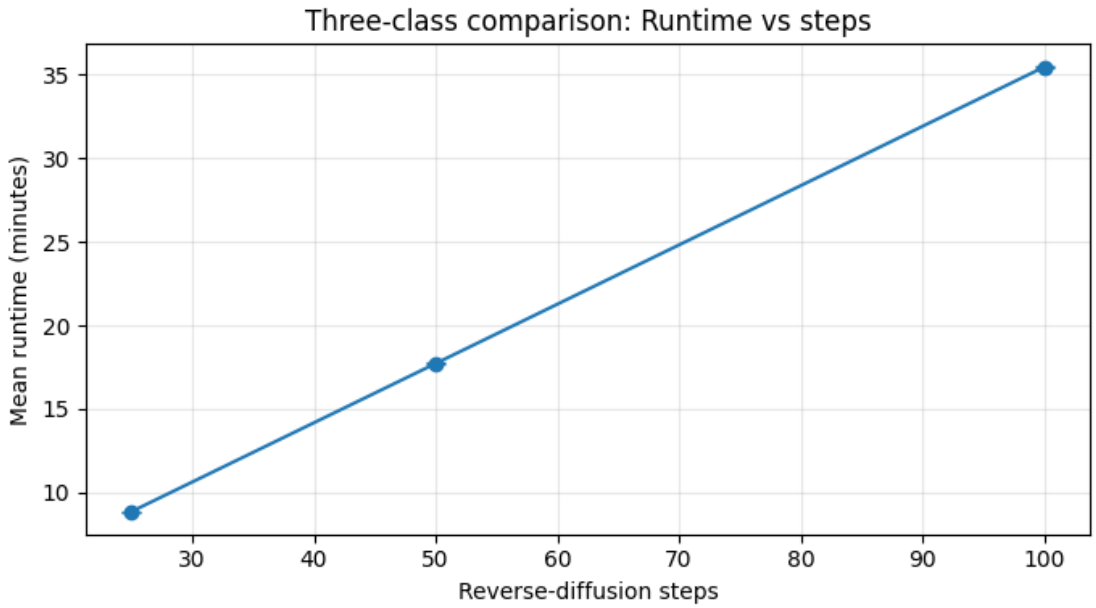


Figure 1. Mean runtime versus reverse-diffusion steps. Runtime falls approximately in proportion to the number of steps.

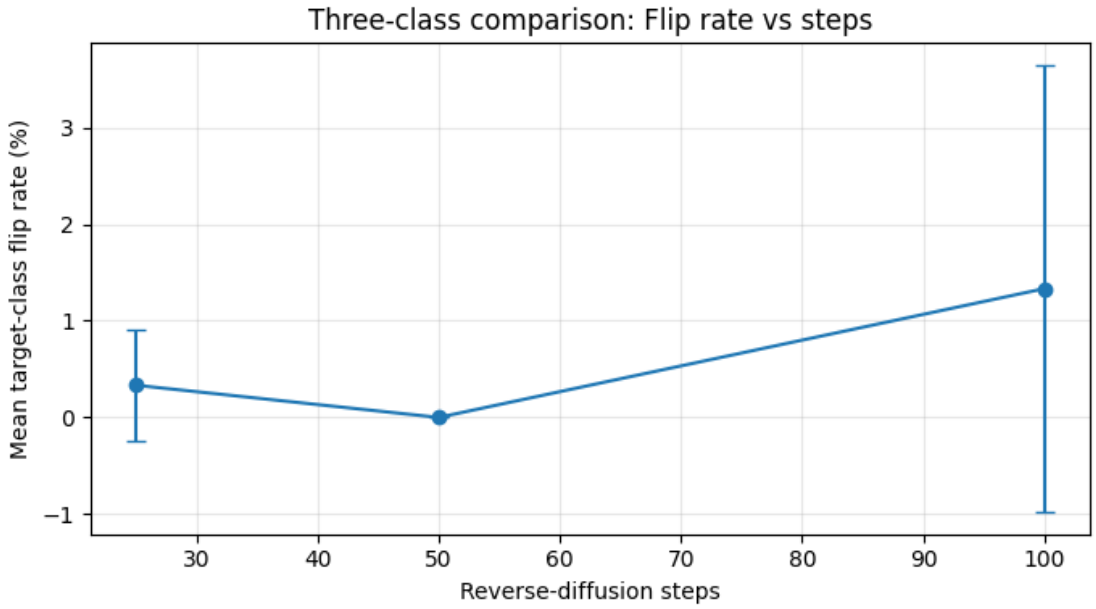


Figure 2. Mean target-class flip rate versus reverse-diffusion steps. Flip rates are near zero for all tested settings.

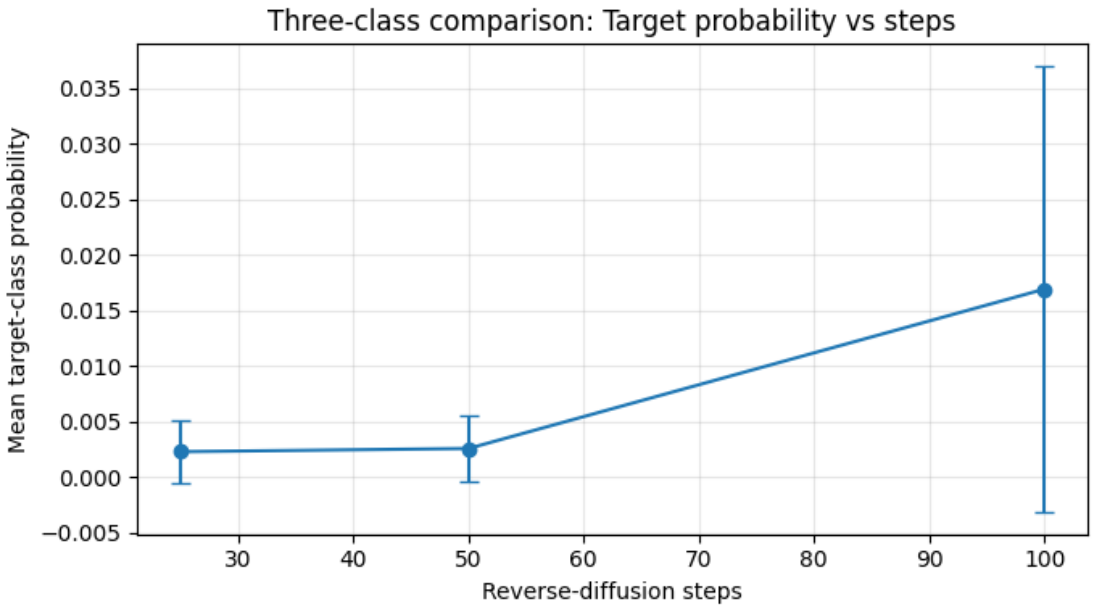


Figure 3. Mean target-class probability versus reverse-diffusion steps. The baseline has the highest mean target probability.

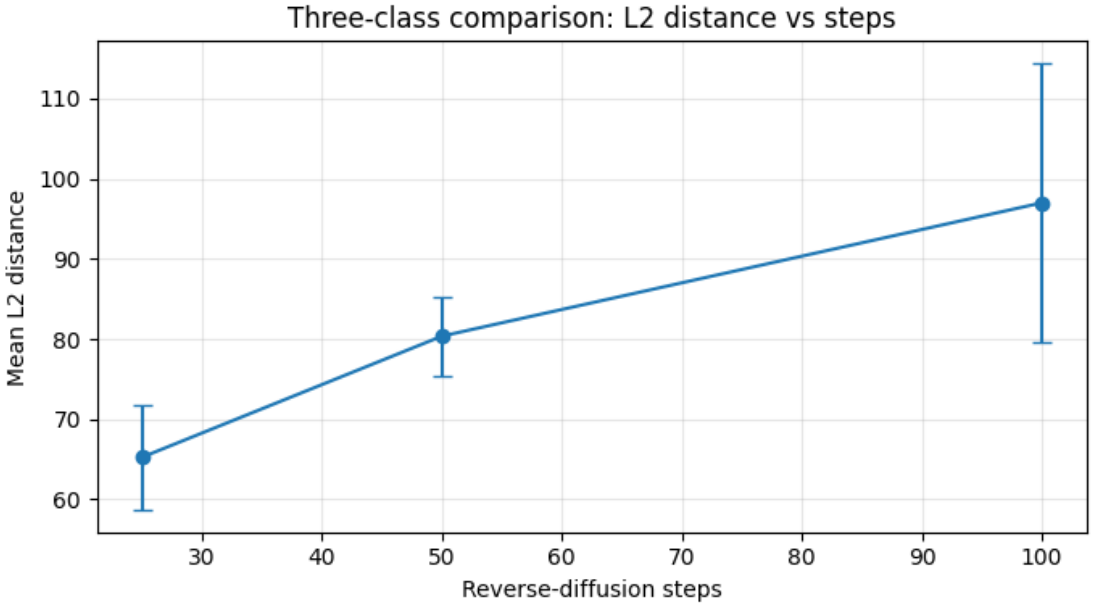
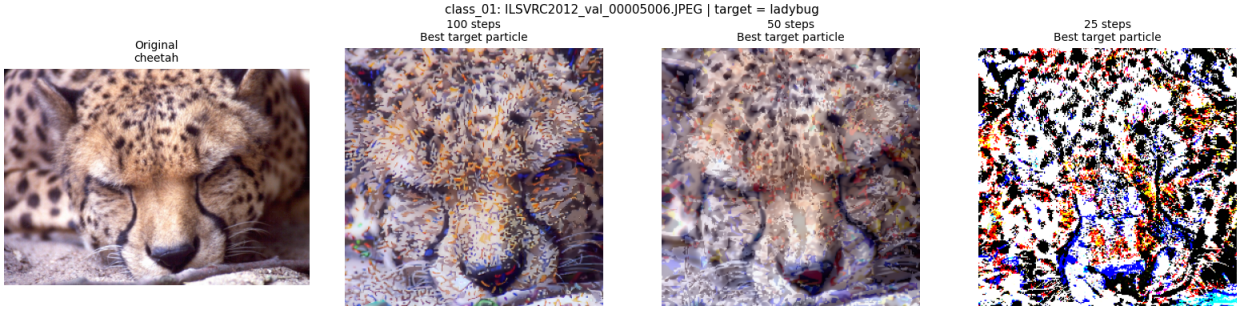


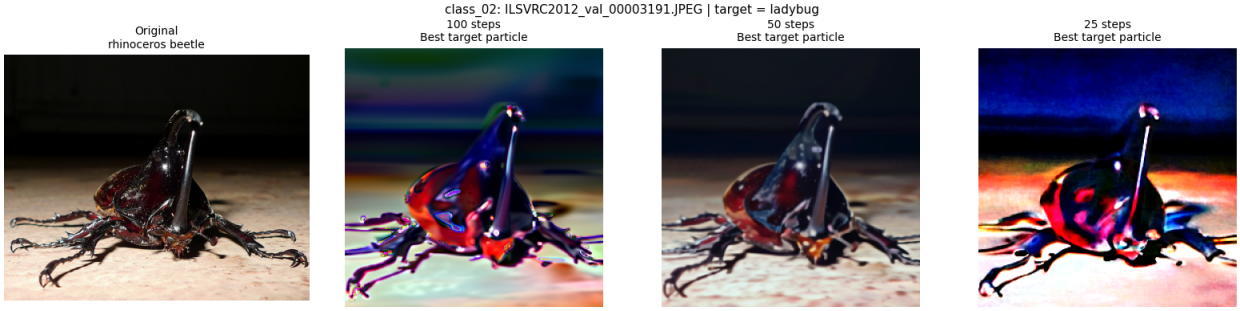
Figure 4. Mean L2 distance versus reverse-diffusion steps. Lower L2 at fewer steps should not be interpreted as improved target success.

8. Counterfactual visualization panels from the notebook

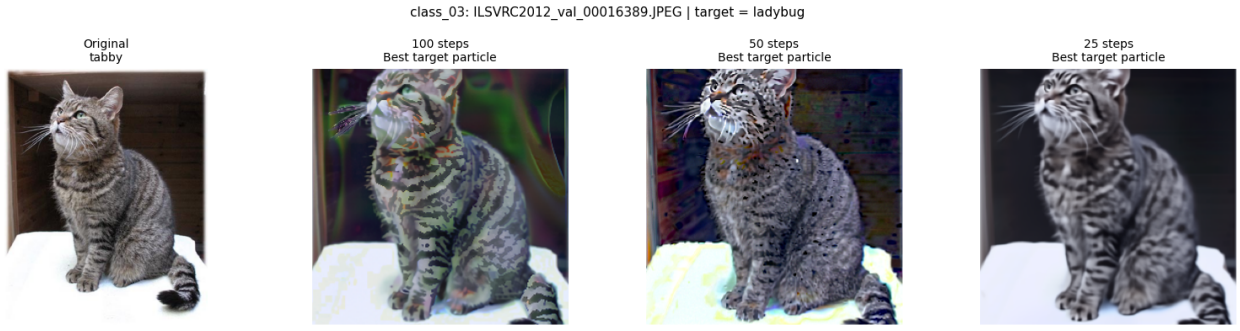
These are the three per-source-class comparison panels emitted by the main experiment cell. They show the notebook's saved intermediate/final visualizations for the 100-, 50- and 25-step runs. The panels should be interpreted together with the quantitative flip-rate and target-probability results; a visual panel alone does not establish that the generated image was classified as ladybug.



Notebook output panel 1 of 3, corresponding to source image 1.



Notebook output panel 2 of 3, corresponding to source image 2.



Notebook output panel 3 of 3, corresponding to source image 3.

9. Supporting research article

Song, Jiaming; Meng, Chenlin; Ermon, Stefano (2021). “Denoising Diffusion Implicit Models.” International Conference on Learning Representations (ICLR 2021). arXiv:2010.02502.

The DDIM paper addresses the computational cost of diffusion sampling. It constructs a family of non-Markovian processes whose reverse sampling can be faster than the standard DDPM chain. The authors report that DDIM can trade computation for sample quality and demonstrate substantially faster sampling in their experiments. This supports the general motivation for investigating fewer reverse-diffusion steps.

How it supports this project: it provides research precedent for studying diffusion inference speed against output quality when the number of sampling steps is reduced. It does not directly validate the particular FreeMCG counterfactual step-count settings tested here; that question is what this project evaluates.

Reference: Song, J., Meng, C., & Ermon, S. (2021). Denoising Diffusion Implicit Models. ICLR 2021. arXiv:2010.02502. https://arxiv.org/abs/2010.02502

10. Interpretation, limitations and conclusion

Conclusion: reducing reverse-diffusion steps produces a clear runtime benefit in this notebook: 50 steps are about twice as fast as 100 steps, and 25 steps are about four times as fast. However, target-class probability drops substantially and target-class flip rates remain close to zero. Therefore, based on these results, 50 and 25 steps cannot be described as preserving counterfactual success. The evidence supports a speed-versus-target-success trade-off, not an unqualified improvement in counterfactual quality.

Limitations:

• Only three images were tested (one per source class).
• All three images use one shared target class (ladybug), so the results do not cover varied source-target pairs.
• One seed/run is reported per image; repeated seeds and more images are needed for stronger conclusions.
• The paper's Table 1 metrics (including human ratings and FID) were not reproduced in this notebook. Our notebook reports runtime, flip rate, target probability and L2 only.
• The paper's 1,000-image protocol and the notebook's 3-image protocol differ, so published and notebook values are not directly comparable.
• No separate Phase 1 result file was supplied in this request. The 100-step condition is used here as the implementation baseline; it should not be represented as an independently re-run Phase 1 result.

11. References

[1] Kim, W. J., Chung, H., Kim, J., Lee, S., Sim, B., & Ye, J. C. (2025). Derivative-Free Diffusion Manifold-Constrained Gradient for Unified XAI. CVPR 2025. Original paper extracts cited by printed page: Figure 4 and reverse-diffusion discussion (p. 6); evaluation protocol (p. 7); Figure 5 and Table 1 (p. 8).

[2] Song, J., Meng, C., & Ermon, S. (2021). Denoising Diffusion Implicit Models. ICLR 2021. arXiv:2010.02502. https://arxiv.org/abs/2010.02502

[3] Project team's executed Phase 2 Jupyter notebook, “Phase 2 — FreeMCG step-count experiment (3 source classes),” cells 13–14, 23–24, 32, 36–40.

Reporting note: All project-specific numbers and plots in Sections 4–8 come from the executed notebook supplied by the user. Published paper numbers are explicitly identified as original-author results.

PHASE 1 RESULTS ADDENDUM

Extracted from the uploaded Phase_1 re-run ZIP. This addendum records the actual saved Phase 1 outputs and is appended to the Phase 2 report.

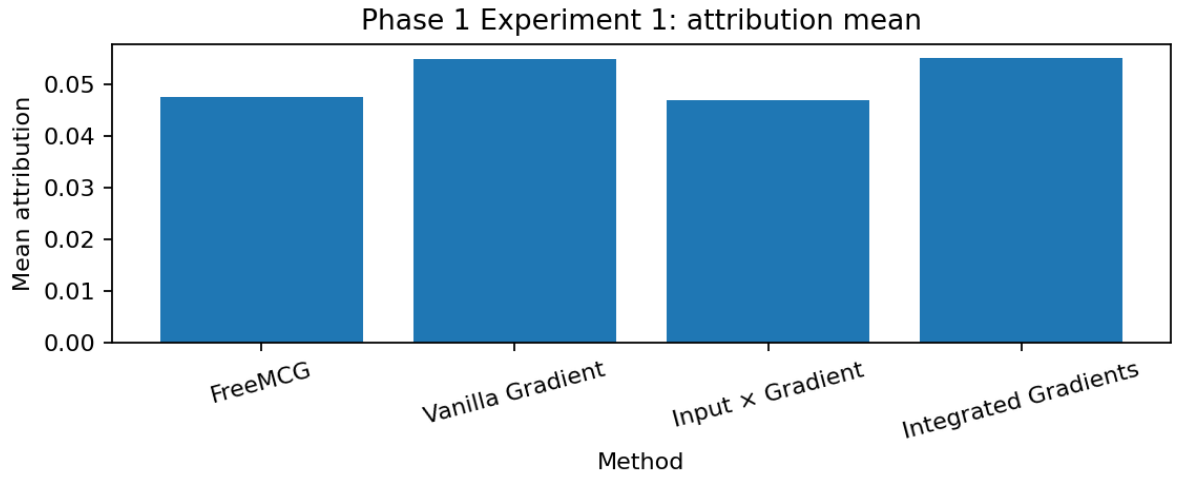
1. Experiment inventory

Experiment	Configuration / output
Feature attribution	ImageNet / ResNet-50 IMAGENET1K_V2; K=10; timestep 400; target rhinoceros beetle (class 306). Attribution CSV and summary J
ROAD evaluation	Same source image and target; 10–50% removal; noise scale 0.05; 12 imputation iterations.
Counterfactual generation	Rhinoceros beetle → ladybug; 100 particles; 100 diffusion steps; strength 0.4; start timestep 400; α=0.2; β=0.02; η=0; seed 0.

2. Experiment 1 — Feature attribution

Saved target class: rhinoceros beetle (class 306); mean target logit 6.4635; mean target probability 0.3275. SHAP status: Not attempted.

Method	Mean	Std	Max	Top 10% mean
FreeMCG	0.04753	0.06953	1.000	0.21466
Vanilla Gradient	0.05479	0.06242	1.000	0.20153
Input × Gradient	0.04680	0.06454	1.000	0.19893
Integrated Gradients	0.05498	0.06636	1.000	0.21071

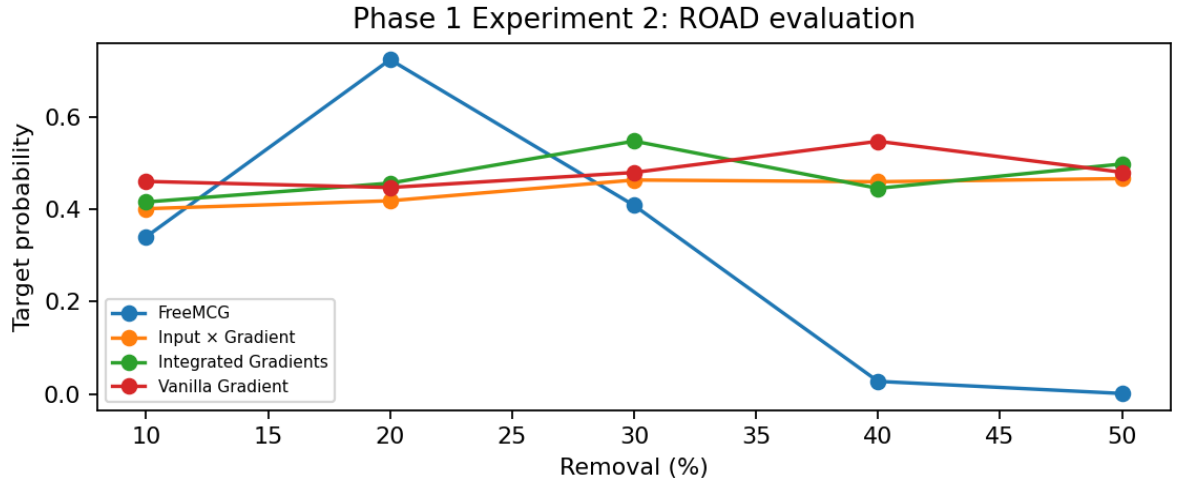


The aggregate attribution statistics are relatively close across the four methods in this one saved run. FreeMCG has the highest top-10%-mean attribution among these four, but these statistics alone do not prove better faithfulness.

3. Experiment 2 — ROAD quantitative evaluation

Original target probability in the summary: 0.409672.

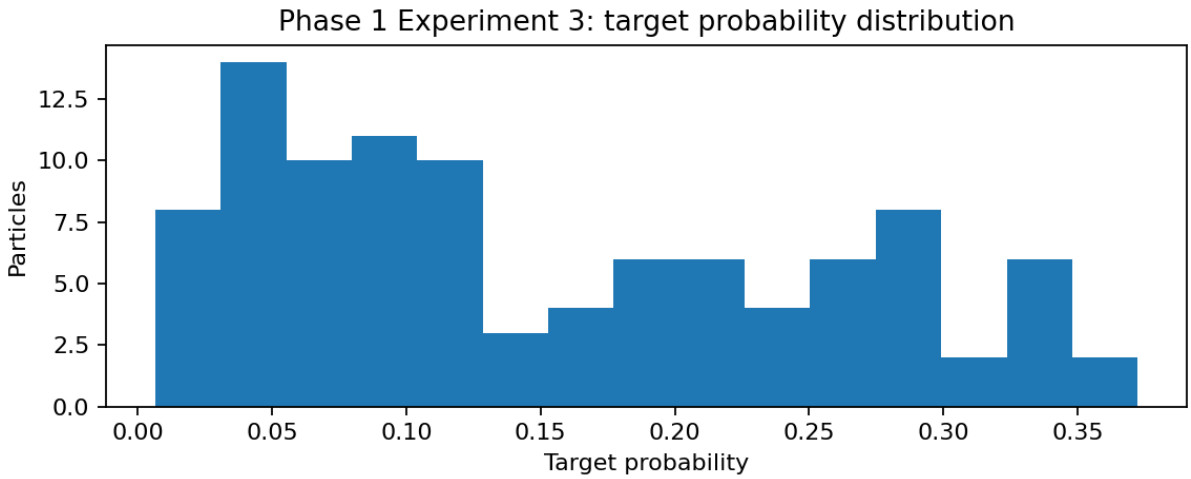
Method	10%	20%	30%	40%	50%
FreeMCG	0.3408	0.7246	0.4087	0.0274	0.0011
Vanilla Gradient	0.4607	0.4475	0.4800	0.5477	0.4805
Input × Gradient	0.4015	0.4187	0.4638	0.4600	0.4668
Integrated Gradients	0.4161	0.4569	0.5481	0.4457	0.4985



At 50% removal, FreeMCG target probability is 0.001090 and predicted class is warplane. Vanilla Gradient, Input × Gradient, and Integrated Gradients retain rhinoceros beetle as predicted class, with probabilities 0.480535, 0.466812, and 0.498479 respectively. FreeMCG's ROAD values are not perfectly monotonic; report the saved values as observed.

4. Experiment 3 — Counterfactual generation

Metric	Saved Phase 1 value
Source-image target probability	0.409672
Ladybug prediction flip rate	55.0% (55/100 particles)
Mean L2 distance	100.832
Median L2 distance	90.120
Best target probability	0.300531
Best particle L2	79.364
Mean particle target probability	0.152786 (calculated from CSV)



The particle CSV contains 100 candidates, of which 55 are predicted as ladybug, matching the saved 55% flip rate. The ZIP also includes original.png, best_counterfactual.png, and difference_map.png.

5. Phase 1 versus Phase 2 — interpretation

Aspect	Phase 1 ZIP	Phase 2 notebook
Source image	Rhinoceros beetle: val_00049168.JPEG	Three: cheetah, rhinoceros beetle val_00003191.JPEG, tabby cat
Target	Ladybug for CF experiment	Ladybug fixed for all three
Particles	100	100
Steps	100	100 / 50 / 25
Seed	0	1 / 2 / 3
Rhino flip rate	55%	4% / 0% / 1%
Rhino mean L2	100.832	80.889 / 84.738 / 71.545

Comparability warning

The Phase 1 and Phase 2 rhinoceros-beetle runs use different ImageNet filenames and different seeds. Thus, their flip-rate and L2 values are not a controlled before/after comparison on the same input. Phase 1 is now documented as a genuine saved 100-step implementation run; Phase 2 remains the within-notebook step-count ablation.

Conclusions and limitations

• Phase 1's 100-step, 100-particle run achieved a 55% ladybug prediction flip rate on its selected image.
• Phase 2's separate rhinoceros-beetle image had flip rates of 4%, 0%, and 1% at 100, 50, and 25 steps.
• Phase 2 runtime fell by about 50% at 50 steps and 75% at 25 steps, but target probabilities also fell in the observed cases.
• Phase 1 experiments are single-image/single-seed saved runs; do not generalize them to all ImageNet images. The ZIP does not include human evaluation or multi-image confidence intervals.

Source files inspected

Experiment 1: attribution_comparison_K10_t400.csv, experiment_1_config.json, experiment_1_summary_K10_t400.json. Experiment 2: experiment_2_road_results.csv, experiment_2_road_50_percent.csv, experiment_2_summary.json. Experiment 3: counterfactual_particle_metrics.csv, experiment_3_summary.json, original.png, best_counterfactual.png, difference_map.png.